

PROGRAM 1

```
public class Main {

    static class Student {
        int roll;
        String name;
        int m1, m2, m3;

        Student(int r, String n, int a, int b, int c) {
            roll = r;
            name = n;
            m1 = a;
            m2 = b;
            m3 = c;
        }

        void result() {
            int total = m1 + m2 + m3;
            double average = total / 3.0;

            System.out.println("Roll No : " + roll);
            System.out.println("Name : " + name);
            System.out.println("Total : " + total);
            System.out.println("Average : " + average);
        }
    }

    public static void main(String[] args) {
        Student s1 = new Student(1, "Hari", 80, 90, 85);
        Student s2 = new Student(2, "Ram", 70, 75, 80);

        s1.result();
        System.out.println();
        s2.result();
    }
}
```

Roll No : 1
Name : Hari
Total : 255
Average : 85.0

Roll No : 2
Name : Ram
Total : 225
Average : 75.0

=== Code Execution Success

PROGRAM 2

```
1 public class Main {
2
3     static class BankAccount {
4         private double balance = 10000;
5
6         void deposit(double amount) {
7             balance += amount;
8         }
9
10        void withdraw(double amount) {
11            if (amount <= balance)
12                balance -= amount;
13            else
14                System.out.println("Insufficient Balance");
15        }
16
17        void showBalance() {
18            System.out.println("Balance = " + balance);
19        }
20    }
21
22    public static void main(String[] args) {
23        BankAccount b = new BankAccount();
24
25        b.deposit(5000);
26        b.withdraw(3000);
27        b.showBalance();
28    }
29 }
```

Balance = 12000.0

=== Code Execution Success

PROGRAM 3

Main.java	Output
<pre>1 class Vehicle { 2 void calculateRent(int days) { 3 } 4 } 5 6 class Car extends Vehicle { 7 void calculateRent(int days) { 8 System.out.println("Car Rent = " + (days * 1000)); 9 } 10 } 11 12 class Bike extends Vehicle { 13 void calculateRent(int days) { 14 System.out.println("Bike Rent = " + (days * 300)); 15 } 16 } 17 18 public class Main { 19 public static void main(String[] args) { 20 Vehicle v = new Car(); 21 v.calculateRent(3); 22 23 v = new Bike(); 24 v.calculateRent(3); 25 } 26 } 27</pre>	<pre>Car Rent = 3000 Bike Rent = 900 === Code Execution Successful ===</pre>

PROGRAM 4

<pre>1 class Employee { 2 String name; 3 double salary; 4 5 Employee(String name) { 6 this.name = name; 7 } 8 9 void calculateSalary() { 10 } 11 } 12 13 class PermanentEmployee extends Employee { 14 PermanentEmployee(String name) { 15 super(name); 16 } 17 18 void calculateSalary() { 19 salary = 50000; 20 System.out.println(name + " Salary = " + salary); 21 } 22 } 23 24 class ContractEmployee extends Employee { 25 ContractEmployee(String name) { 26 super(name); 27 } 28 29 void calculateSalary() { 30 salary = 25000; 31 System.out.println(name + " Salary = " + salary); 32 } 33 } 34 35 class Main { 36 public static void main(String[] args) { 37 Employee e1 = new PermanentEmployee("Hari"); 38 Employee e2 = new ContractEmployee("Ram"); 39 40 e1.calculateSalary(); 41 e2.calculateSalary(); 42 } 43 } 44</pre>	<pre>Hari Salary = 50000.0 Ram Salary = 25000.0 === Code Execution Successful ===</pre>
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PROGRAM 5

main.java	Output
<pre>1 class Book { 2 private int bookId; 3 private String title; 4 private String author; 5 private boolean issued; 6 7 Book(int id, String t, String a) { 8 bookId = id; 9 title = t; 10 author = a; 11 issued = false; 12 } 13 14 void issueBook() { 15 if (!issued) { 16 issued = true; 17 System.out.println("Book Issued"); 18 } else { 19 System.out.println("Already Issued"); 20 } 21 } 22 23 void returnBook() { 24 if (issued) { 25 issued = false; 26 System.out.println("Book Returned"); 27 } else { 28 System.out.println("Book was not Issued"); 29 } 30 } 31 32 void display() { 33 System.out.println(bookId + " " + title + " " + author + " Issued: " + issued); 34 } 35 } 36 37 class Main { 38 public static void main(String[] args) { 39 Book b = new Book(101, "Java", "James"); 40 41 b.display(); 42 b.issueBook(); 43 b.display(); 44 b.returnBook(); 45 b.display(); 46 } 47 } 48</pre>	<pre>101 Java James Issued: false Book Issued 101 Java James Issued: true Book Returned 101 Java James Issued: false === Code Execution Successful ===</pre>